Ozgur Ural, Ph.D.

Last update on August 9, 2026

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Machine Learning Research Scientist & Senior Software Engineer, Ph.D.

Machine learning research scientist and senior software engineer with 12+ years building production systems across cybersecurity, defense, and simulation, now leading enterprise AI strategy and applied GenAI at Avion. Completed a Ph.D. in Electrical Engineering and Computer Science at Embry-Riddle Aeronautical University in the United States (August 2025), with a machine-learning dissertation and three first-author IEEE Access publications on Proof-of-Learning and model watermarking. My work centers on **verifiable, trustworthy machine learning**: systems whose behavior can be proven rather than assumed, engineered for domains where failure is not an option.

Deep foundation in systems programming (C, C++, Scala) and distributed architecture, combined with hands-on ML-security research. Experienced in leading engineering teams, owning complex technical deliverables end to end, and translating original research into production-grade systems.

TECHNICAL SKILLS

- **Languages:** C, C++, Scala, Python, Go, Java, TypeScript, JavaScript
- **Systems and real-time:** multithreading, async I/O, low-latency, backpressure, Linux, networking, Boost, Qt/QML
- **Distributed and data:** gRPC, Protobuf, Redis, PostgreSQL, REST, real-time data pipelines
- **Machine learning & AI:** PyTorch, TensorFlow, scikit-learn, deep learning, LLMs, retrieval-augmented generation (RAG), AI agents, on-prem and air-gapped model deployment, federated and distributed ML training, large-scale model evaluation; trustworthy and verifiable ML (model watermarking, Proof-of-Learning, adversarial robustness, model provenance)
- **Web and UI:** Svelte, TypeScript, HTML, CSS, Tailwind
- **DevOps:** Docker, GitLab CI/CD, Jenkins, Git

EXPERIENCE

Avion Full Flight Simulators — Nieuw-Vennep, Netherlands

Senior Software Engineer — Oct 2023 – Present

- Lead Avion's enterprise AI strategy: authored the AI adoption roadmap and a three-tier architecture spanning air-gapped on-prem LLMs, a governed cloud-LLM API tier, and edge deployment.
- Designed a suite of retrieval-augmented and autonomous AI agents (engineering-knowledge copilot with grounded citations and change-impact analysis, autonomous repository agent, RFP accelerator, training-debrief writer) and built the strategy portal end to end (Flask, SQLite, Docker, GitLab CI/CD).
- Developed real-time simulator software in C, C++, and Scala.
- Built a low-latency data pipeline ingesting simulator telemetry, normalizing and serializing to JSON and Protobuf, caching in Redis, persisting in PostgreSQL, and serving the frontend via gRPC and HTTP; sustained approximately 50 GB/s real-time throughput using batching and bounded-queue backpressure.
- Built web applications for the Avion Cloud Environment to monitor, configure, and diagnose core simulator components using TypeScript, Svelte, Scala, Python, and gRPC.
- Containerized and deployed services with Docker; automated testing and deployments via GitLab CI/CD.
- Applied multithreading and async I/O patterns with backpressure to meet rigorous real-time requirements.

Embry-Riddle Aeronautical University, Daytona Beach — Florida, USA

Graduate Research Assistant — Sep 2021 – Oct 2025

- Advisor: Dr. Kenji Yoshigoe. Dissertation: Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking.
- Conducted original research on verifiable, trustworthy machine learning across proof-of-learning, model watermarking, and adversarial robustness, with three first-author papers in IEEE Access (2025 JCR impact factor 4.2).
- Journal referee for IEEE Access; program committee member for NLPAICS 2026 (University of Alicante).

Havelsan — Ankara, Turkey

Software Team Lead — Nov 2020 – Aug 2021

Havelsan DLP enterprise Data Leakage Prevention product.

- Led a team of 14 engineers (developers, QA, DevOps, support) delivering endpoint security software.
- Developed and maintained core features of the endpoint agent in C, C++, Java, and Go.
- Coordinated with product and business stakeholders to align delivery with roadmap objectives.

STM Defence Technologies ANKARA, TURKEY

Expert Software Engineer Feb 2019 – Nov 2020

Kargu Autonomous Rotary-Wing Attack UAV and Togan Autonomous Multi-Rotor Reconnaissance UAV.

- Developed critical components of the mission-control and ground-control systems in C++11 with Boost and Qt, under hard-real-time, safety-critical constraints.
- Designed and implemented Power-On and Continuous Built-In Tests for system reliability validation.
- Worked with cross-functional teams to meet military-grade engineering and safety standards.

Comodo Cybersecurity ANKARA, TURKEY

Expert Software Engineer Jul 2014 – Feb 2019

Comodo Dome Secure Web Gateway, Comodo Patch Manager, and the Comodo Dragon browser.

- Led design and development of Windows services and applications in C++17 with Boost and WinAPI.
- Established unit testing and code-review practices; implemented CI/CD pipelines with Jenkins.
- Developed statistical reporting modules and advanced browser security features.

EDUCATION

Embry-Riddle Aeronautical University, Daytona Beach FLORIDA, USA

Ph.D. in Electrical Engineering and Computer Science Aug 2021 – Aug 2025

- Dissertation: Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Advisor: Dr. Kenji Yoshigoe; committee included IEEE Fellow Dr. Houbing Song.

Middle East Technical University (METU) ANKARA, TURKEY

Master of Science in Cyber Security 2015 – 2019

- Thesis: Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. Advisor: Prof. Cengiz Acartürk.

Middle East Technical University (METU) ANKARA, TURKEY

Bachelor of Science in Computer Engineering 2010 – 2014

- Institutional note: METU is widely regarded as Turkey's leading engineering university and is ranked 269th globally (QS World University Rankings 2026).

Anadolu University ESKISEHIR, TURKEY

Bachelor of Business Administration 2012 – 2017

PUBLICATIONS

- Ural, O. and Yoshigoe, K. (2025). SecurePoL: Integration of Watermarking With Proof-of-Learning to Enhance Security Against Spoofing Attacks. *IEEE Access*, vol. 13, pp. 213067–213091.
- Ural, O. and Yoshigoe, K. (2024). Enhancing Security of Proof-of-Learning against Spoofing Attacks using Feature-Based Model Watermarking. *IEEE Access*.
- Ural, O. and Yoshigoe, K. (2023). Survey on Blockchain-Enhanced Machine Learning. *IEEE Access*, pp. 145331–145362.
- Ural, O. and Acartürk, C. (2021). Automatic Detection of Cyber Security Events from Turkish Twitter Stream and Newspaper Data. ICISSP 2021.
- Ural, O. and Erdur, E. (2016). Secure Proxy on Cloud. Preprint. DOI: 10.13140/RG.2.2.24058.08649.
- Ural, O. (2014). Autonomous Cargo and Mail Delivery. Turkish Autonomous Robots Conference, Ankara, Turkey.

- Ural, O. (2025). Enhancing Proof-of-Learning Security Against Spoofing Attacks Using Model Watermarking. Doctoral dissertation, Embry-Riddle Aeronautical University.

For full list and citation metrics: Google Scholar profile.

PROFESSIONAL SERVICE

- **Journal Referee:** IEEE Access and peer-reviewed journals in security, machine learning, distributed systems, and blockchain.
- **Program Committee Member (2026):** 2nd Workshop on NLP Applied to Information and Cyber Security (NLPAICS 2026), University of Alicante, Spain.
- **Reviewer:** IEEE and ACM conferences in security, machine learning, and distributed systems.

AWARDS AND HONORS

- First Prize, TUBITAK National Project Competition (Turkey's national science and technology council), June 2014.
- Second Prize, METU Computer Engineering Graduation Projects (Demo Day), June 2014.